

★ CONDUCTION + CONVECTION ★

GS | Ch02 Physics B | Topic 1 of 12

► Meaning in one line

Solid-touch vs fluid-flow , examples.

► Key points

Point	Example
Conduction: solids + touch + metals best.	spoon heats.
Convection: fluids + bulk flow.	sea/land breeze + radiator.

★ IMPORTANT FOR EXAM

Radiation + expansion

⚠ TRAP - read twice

Mode-only + 'heat moves' + medium-miss + example-miss

✓ **Try yourself:** Medium + flow + example + transfer + sort? **Answer:** Solid vs fluid.

★ RADIATION + EXPANSION ★

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► Meaning in one line

No-medium waves , solids/liquids/gases expand.

► Key points

Point	Example
Radiation: no medium + dark absorbs.	sun + cooker.
Expansion: heat expands + gaps/joints.	rails + bridges + bimetal.

★ IMPORTANT FOR EXAM

Specific heat + change of state

⚠ TRAP - read twice

Heat-only + 'expands' + medium-miss + direction-miss

✓ **Try yourself:** Waves + gaps + medium + effect? **Answer:** Radiation + expand.

★ SPECIFIC HEAT + LATENT HEAT ★

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► Meaning in one line

Capacity , hidden heat , water high.

► Key points

Point	Example
Specific heat: $Q = m c \text{ change } T$ + water highest.	sea moderates.
Latent: flat at melt/boil + hidden.	steam burn worse.

★ IMPORTANT FOR EXAM

Reflection + mirrors

⚠ TRAP - read twice

Temp-only + 'heats' + flat-miss + capacity-miss

✓ **Try yourself:** Capacity + flat + water + heat? **Answer:** c + latent.

★ REFLECTION + PLANE MIRROR ★

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► Meaning in one line

Laws , virtual image , lateral.

► Key points

Point	Example
Laws: angle in = angle out + same plane.	-
Plane: virtual + erect + same size/distance + lateral flip.	periscope pair.

★ IMPORTANT FOR EXAM

Spherical + refraction

⚠ TRAP - read twice

Real-only + 'image asked' + virtual-miss + flip-miss

✓ **Try yourself:** Angle + virtual + flip + mirror? **Answer:** Laws + image.

★ SPHERICAL MIRRORS + REFRACTION ★

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► Meaning in one line

Concave/convex , bend at boundary.

► Key points

Point	Example
Concave converges + convex diverges.	torch + rear view.
Refraction: speed changes + bends.	straw bent + pool shallow.

★ IMPORTANT FOR EXAM

Lenses + eye

⚠ TRAP - read twice

Mirror-only + 'bend asked' + converge-miss + medium-miss

✓ **Try yourself:** Curve + medium + bend + use? **Answer:** Mirror + refract.

★ LENSES + HUMAN EYE ★

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► Meaning in one line

Convex/concave , retina/lens/ciliary.

► Key points

Point	Example
Convex converges + concave diverges.	magnifier + specs.
Eye: lens focuses + retina senses + ciliary adjusts.	real inverted image.

★ IMPORTANT FOR EXAM

Defects + instruments

⚠ TRAP - read twice

Glass-only + 'lens asked' + converge-miss + retina-miss

✓ **Try yourself:** Converge + retina + adjust + focus? **Answer:** Lens + eye.

★ DEFECTS + OPTICAL INSTRUMENTS ★

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► Meaning in one line

Myopia/hypermétropia , concave/convex fix , scope.

► Key points

Point	Example
Myopia: far blur + concave. Hyper: near blur + convex.	-
Instruments: magnify + lenses/mirrors.	lab + stars.

★ IMPORTANT FOR EXAM

Sound + frequency

⚠ TRAP - read twice

Blur-only + 'vision asked' + lens-miss + near/far-miss

✓ **Try yourself:** Near + far + lens + defect + fix? **Answer:** Defect + lens.

★ SOUND + FREQUENCY/PITCH ★

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► Meaning in one line

Mechanical wave , f , pitch link.

► Key points

Point	Example
Sound: mechanical + medium must.	no sound in vacuum.
Frequency high = shrill + low = grave.	woman vs man voice.

★ IMPORTANT FOR EXAM

Loudness + wave relation

⚠ TRAP - read twice

Air-only + 'sound asked' + medium-miss + pitch-miss

✓ **Try yourself:** Medium + pitch + wave + sound? **Answer:** Medium + f.

★ LOUDNESS + WAVE RELATION ★

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► Meaning in one line

Amplitude, $v = f \lambda$.

► Key points

Point	Example
Amplitude big = loud + small = soft.	whisper vs shout.
$v = f \times \lambda$ + units.	double f = half λ .

★ IMPORTANT FOR EXAM

Ranges + ultrasound

⚠ TRAP - read twice

Loud-only + 'wave asked' + amplitude-miss + product-miss

✓ **Try yourself:** Swing + product + units + wave? **Answer:** Amplitude + $v = f \lambda$.

★ RANGES + ULTRASOUND USES ★

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► Meaning in one line

Infrasonic/audible/ultrasonic , scan/clean.

► Key points

Point	Example
Infrasonic below 20 + audible 20-20000 + ultrasonic above.	-
Ultrasound: scan/clean/sonar.	pregnancy scan + ships.

★ IMPORTANT FOR EXAM

Echo + charge/current

⚠ TRAP - read twice

Hear-only + 'sound asked' + range-miss + use-miss

✓ **Try yourself:** Range + use + Hz + scan? **Answer:** Range + uses.

★ ECHO + CHARGE/CURRENT ★

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► Meaning in one line

Reflect , $Q = I t$.

► Key points

Point	Example
Echo: reflection + 17 m min.	sonar + hills.
Charge $Q = I \times t$ + coulomb + current flow.	battery drain.

★ IMPORTANT FOR EXAM

Voltage + Ohm law

⚠ TRAP - read twice

Sound-only + 'echo asked' + distance-miss + flow-miss

✓ **Try yourself:** Reflect + flow + distance + charge? **Answer:** Echo + $Q=It$.

★ VOLTAGE + OHM LAW ★

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► Meaning in one line

Push , $V = I R$.

► Key points

Point	Example
Voltage: push + volt.	cell + mains.
Ohm: $V = I R$ + ohm.	double V = double I .

★ IMPORTANT FOR EXAM

Till end: series/parallel + power!

⚠ TRAP - read twice

Current-only + 'circuit asked' + push-miss + ratio-miss

✓ **Try yourself:** Push + ratio + volt + ohm? **Answer:** $V = I R$.